

**DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING
NORTH CAROLINA STATE UNIVERSITY**

CHE 330

Homework Set 4

Problem 1 (20%) A company claims that using their specialty vacuum will “significantly reduce” the time needed to vacuum a 2000 ft² home. Assume it is well-established that it takes 45 minutes to vacuum a 2000 ft² home, with a standard deviation of 4.2 minutes. Utilizing the specialty vacuum in 9 separate tests demonstrates that the specialty vacuum takes an average of 41.5 minutes to vacuum a 2000 ft² home. Is the company’s claim justified at the **(a, 10%)** 95% confidence level? **(b, 10%)** 99% confidence level?

Problem 2 (30%) You are a quality control engineer at a popcorn company claiming that its bags of microwaveable popcorn contain a mean of 1250 kernels with a standard deviation of 50 kernels. A consumer watchdog group has written you a letter claiming that your company is not including the appropriate number of kernels of popcorn per bag. As evidence, they send you results showing that a random sample of 16 bags had a mean of 1226 kernels per bag. Using their data, how would you phrase your research question if you want the null hypothesis to be justified at the 95% confidence level? Show calculations comparing results for a one tail question and a two tail question.

Problem 3 (30%) A local pizzeria offers a “Pepperoni Feast” pizza advertised to be covered with (on average) 100 slices of pepperoni. You’ve had some time on your hands lately, so you’ve counted the number of slices of pepperoni on each of the nine Pepperoni Feast pizzas you’ve ordered in the past few months. You’ve collected the following series of results: 97, 105, 90, 92, 89, 92, 97, 105, 94. Based on your data, can the pizzeria’s claim that Pepperoni Feast pizzas contain an average of 100 slices of pepperoni be disproved at the **(a, 15%)** 95% confidence level? **(b, 15%)** 99% confidence level?

Problem 4 (20%) It is well-established that systolic blood pressure for the US population is a normally-distributed variable centered around a mean of 120 mm_{Hg} with a standard deviation of 12 mm_{Hg}. A researcher studying inhabitants of a small town in rural North Carolina suspects the town’s residents have a significantly lower systolic blood pressure than the US population. Data collected from a random sample of 9 residents from the rural town demonstrates a mean systolic blood pressure of 111 mm_{Hg}. Based on this data, can it be stated that the residents of the rural town have a mean systolic blood pressure significantly lower than the US population at the **(a, 10%)** 95% confidence level? **(b, 10%)** 99% confidence level?